Netflix Data Analysis Project

Project Overview

Performed end-to-end data analysis on Netflix dataset using SQL and Power BI. Focused on data cleaning, transformation, and extracting actionable insights to understand content trends, distribution, and growth patterns.

Tools & Technologies

1. MySQL (Data Cleaning & Analysis)
2. Power BI (Dashboard & Visualization)

Highlights

1. Analyzed 8000+ Netflix records using advanced SQL queries
2. Implemented data cleaning pipeline improving data quality
3. Built interactive Power BI dashboard for business insights
4. Used window functions and subqueries for complex analysis
5. Identified content trends and growth opportunities

Data Cleaning & Preparation

1. Checked duplicate records using GROUP BY and HAVING
2. Standardized categorical values (Movie / TV Show)
3. Handled missing values by replacing with NULL or defaults
4. Converted inconsistent date formats into standard DATE format
5. Split duration column into minutes and seasons
6. Created new primary key column for uniqueness

Key Analysis Performed

1. Content distribution between Movies and TV Shows
2. Top directors and content contribution
3. Genre popularity analysis
4. Country-wise content production
5. Rating distribution
6. Year-wise growth trends
7. Top 3 shows per genre using window functions

8 Average duration analysis by country

Key Insights

1. Movies dominate Netflix catalog compared to TV Shows.

type	total_count
Movie	4712
TV Show	3288

2. Drama ,Sci-Fi and Comedy are most frequent genres.

listed_in	Total_Count
Comedy	1050
Sci-Fi	1022
Drama	1022
Documentary	1020
Action	988
Romance	983
Thriller	971
Horror	944

3. France ,USA and India contribute highest content.

CountryName	Total_Count
France	938
United States	907
India	889
Australia	885
Canada	881
UK	868
Germany	846

4. Content production increased significantly after 2015.

Release_year	total_content_count
2015	73
2016	91
2017	85
2018	87
2019	96
2020	86
2021	93
2022	78
2023	70
2024	77
2025	99

5. Certain directors dominate platform content.

director	total_no_of_Movies
Director 118	13
Director 359	12
Director 410	12
Director 1	12
Director 355	12
Director 354	11
Director 31	11
Director 494	11
Director 91	11
Director 123	11
Director 277	11
Director 67	11

director	total_no_of_TVShows
Director 476	9
Director 85	8
Director 443	8
Director 487	8
Director 321	7
Director 319	7
Director 318	7
Director 63	7
Director 329	7
Director 80	7
Director 490	7
Director 108	7

6. Viewer preference leans towards short duration movies.

duration_category	total_movies
Long	490
Mid-duration	156
Short	937

7. Content ratings shows that the majority of titles fall under PG, TV-14, and TV-MA categories.

Rating_Category	total_count
PG	1172
TV-14	1167
TV-MA	1141
PG-13	1119
R	1099

Business Recommendations

- Invest more in high-demand genres like Comedy and Drama.
- Expand content production in top-performing countries.
- Focus on TV Shows as binge-watching is increasing.
- Improve content availability across regions.

Conclusion

This project demonstrates the ability to handle raw data, perform data cleaning, analyze trends, and build an interactive dashboard. It provides actionable insights that can help Netflix improve its content strategy.